

Student Computer Science Project

Patterns formation in Sandpile on Square Lattice

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Goal

Study the Abelian sandpile model on a square lattice. Label sites by integer coordinates (i, j) , define what it means for (i, j) to be unstable, and let unstable sites relax by sending grains to their nearest neighbors (i', j') in one time step. Run this rule on a computer and look at the final patterns. Do diverse experiments.

Lattice and heights

Consider a large finite square lattice

$$\{1, 2, \dots, L\} \times \{1, 2, \dots, L\},$$

with sites labeled (i, j) . At each site (i, j) add some grains $\eta(i, j) \geq 0$. Each site (i, j) has up to four nearest neighbors:

$$(i+1, j), \quad (i-1, j), \quad (i, j+1), \quad (i, j-1).$$

If a neighbor falls outside the grid, treat it as a boundary, and any grain going there is lost. The lattice boundaries are open mean grains can leave the system (sink).

Unstable sites and relaxation

Fix relaxation rule: a site (i, j) is called *unstable* if

$$\eta(i, j) \geq 4.$$

One relaxation (one time step) at an unstable site (i, j) does the following:

$$\begin{aligned} \eta(i, j) &\rightarrow \eta(i, j) - 4, \\ \eta(i+1, j) &\rightarrow \eta(i+1, j) + 1 \quad \text{if } (i+1, j) \text{ is inside the grid,} \\ \eta(i-1, j) &\rightarrow \eta(i-1, j) + 1 \quad \text{if } (i-1, j) \text{ is inside the grid,} \\ \eta(i, j+1) &\rightarrow \eta(i, j+1) + 1 \quad \text{if } (i, j+1) \text{ is inside the grid,} \\ \eta(i, j-1) &\rightarrow \eta(i, j-1) + 1 \quad \text{if } (i, j-1) \text{ is inside the grid.} \end{aligned}$$

So in one step, (i, j) sends one grain to each nearest neighbour (i', j') and loses four grains. If a neighbour is outside, that grain disappears.

This relaxation may cause further sites to be unstable and thus, process will continue until every site (i, j) is stable, i.e. $\eta(i, j) \leq 3$ for all (i, j) .

Driving (adding grains)

Choose central driving rule: Let $(i_0, j_0) = (\lfloor L/2 \rfloor, \lfloor L/2 \rfloor)$. Start from the empty configuration

$$\eta(i, j) = 0 \quad \text{for all } (i, j).$$

Then repeat the following M times (it implies the addition of M grains to the lattice):

1. Add one grain at the central site (i_0, j_0) :

$$\eta(i_0, j_0) \rightarrow \eta(i_0, j_0) + 1.$$

2. While there exists an unstable site (i, j) with $\eta(i, j) \geq 4$, perform relaxations at such sites.
3. Repeat the step 1.

After adding grains, say M (very large), observe at the final stable configuration.

Other simple choices (optional)

Instead of always adding at the centre, can try also:

- Add at a fixed site (i_1, j_1) near the boundary.
- Choose a site (i, j) at random in a small square around the centre, and add one grain there.

In all the case case, keep the relaxation rule same.

What to measure

Consider the following observables:

- **Final picture.** Plot a picture of the final heights $\eta(i, j)$, using four colours for heights 0, 1, 2, 3.
- **Height counts.** Count how many sites have height 0, 1, 2, 3.
- **Toppled region.** Mark all sites (i, j) that ever became unstable at least once.
- **Radius.** Measure how far the toppled region spreads from the centre.

Repeat this for different values of L and M and compare the pictures.

Deliverables

- A short report (4-6 pages) explaining:
 - The introduction of sandpile model, brief literature review and other sandpile based model
 - Detailed driving and relaxation rules.
 - Pictures of final emergent pattern and simple measurements.
- Code that:
 - Stores $\eta(i, j)$ in an array,
 - Finds unstable sites and relaxes them,
 - Repeats add-and-relax many times,
 - Produces images of $\eta(i, j)$.

References

1. D. Dhar, T. Sadhu, and S. Chandra, Pattern formation in growing sandpiles <https://iopscience.iop.org/article/10.1209/0295-5075/85/48002>.
2. D. Dhar and T. Sadhu, A sandpile model for proportionate growth, <https://iopscience.iop.org/article/10.1088/1742-5468/2013/11/P11006>.